Prob 1.)

- A microcanonical ensemble has energy, size, and species amount fixed.

(natural variables)

E → internal energy for this ensemble

M → size of ensemble

N → species amount for this ensemble.

a.) $M = \sum_{i=1}^{N} \sigma_i M$ , $\sigma_i = \pm 1$

$\sigma_+ = 1$ (spin up)

$\sigma_- = -1$ (spin down)

- $BH(E, N, M)$

- $M = \sum_{i=1}^{N} \sigma_i M = M(\sigma_+ - \sigma_-)$

- $\sigma_+ + \sigma_- = N$ $\left(\sigma_- = N - \sigma_+\right)$

$M = M(2\sigma_+ - N)$

- $E = -H_M(2\sigma_+ - N)$

$$E = -HM(2\sigma_+ - N)$$

$$\beta = \left(\frac{\partial \ln(\Omega(E))}{\partial E}\right)_{N,M}$$

$$P = \frac{1}{\Omega}$$

$$\beta = \frac{1}{k_B T}$$

$$H = \frac{-E}{M(2\sigma_+ - N)}$$

$$\beta H = \frac{-E}{k_B T M(2\sigma_+ - N)}$$

$$\Omega(E) = \frac{N!}{\sigma_+!(N-\sigma_+)!}$$

(Total amount of Spins up to choose from total N)

$$\frac{\partial \ln(\Omega(E))}{\partial E} = \frac{\partial \ln(\Omega)}{\partial \sigma^+} \frac{\partial \sigma^+}{\partial E}$$

$$\ln(\Omega) = \ln(N!) - \ln(\sigma_+!(N-\sigma_+)!)$$

$$\cdot \frac{\partial \ln(\Omega)}{\partial \sigma^+} = \frac{-\partial}{\partial \sigma^+}\left(\ln\left[\sigma_+!\,(N-\sigma_+)!\right]\right)$$

$$\frac{\partial E}{\partial \sigma^+} = -2HM \quad , \quad \frac{\partial \sigma^+}{\partial E} = \frac{-1}{2HM}$$

$$\beta = \frac{\partial \ln(\Omega)}{\partial E} = \frac{-\partial}{\partial \sigma^+}\left(\ln\left[\sigma_+!\,(N-\sigma_+)!\right]\right)\left(\frac{1}{2HM}\right)$$

Stirling approx:

$$\ln(n!) \approx n\ln(n) - n$$

$$\cdot \ln\left[\sigma_+!\,(N-\sigma_+)!\right] =$$

$$\cdot \ln(\sigma_+!) + \ln((N-\sigma_+)!)$$

· applying stirling approx:

$$\cdot \ln\left[\sigma_+!\,(N-\sigma_+)!\right] \approx \sigma_+\ln(\sigma_+) - \sigma_+ + (N-\sigma_+)\ln(N-\sigma_+) - (N-\sigma_+)$$

$$\approx \sigma_+\ln(\sigma_+) + (N-\sigma_+)\ln(N-\sigma_+) - 2\sigma_+ - N$$

$$H\beta = \frac{1}{2M}\frac{\partial}{\partial \sigma_+}\left(\ln\left[\sigma!\,(N-\sigma_+)!\right]\right)$$

## $S(\beta, H, N)$:

- $S = k_B \ln(\Omega)$

$$\ln(\Omega) = \ln(N!) - \ln(\sigma_+!) + \ln((N-\sigma_+)!)$$

apply Stirling approx ↘

$$\approx N\ln(N) - \sigma_+ \ln(\sigma_+) + (N-\sigma_+)\ln((N-\sigma_+)!)$$

$$N - \sigma_+ = \sigma_-$$

$$\ln(\Omega) = N\ln(N) - \sigma_+\ln(\sigma_+) + \sigma_-\ln(\sigma_-)$$

$$S = k_B N \ln(N) - k_B \sigma_+ \ln(\sigma_+) + k_B \sigma_- \ln(\sigma_-)$$

b.)

- Prove that $P(\sigma) = Ce^{-\beta E(\sigma)}$
- $\epsilon(\sigma) = -H\mu\sigma$ .

- $P = \frac{1}{\Omega}$ (Microcanonical ensemble)

$$\beta = \frac{\partial \ln(\Omega)}{\partial E} = \frac{1}{2\mu H} \frac{\partial}{\partial \sigma_+} \left( \ln\left[ \sigma_+! (N-\sigma_-)! \right] \right)$$

- $\frac{\partial \ln(\Omega)}{\partial E} = \beta$
- $E_n = -\mu\sigma H$ (1 energy unit)
- $\int \partial \ln(\Omega) = \int \beta \partial E$
- $\ln(\Omega) = \beta E + C$
- $\Omega = Ce^{\beta E}$

$$\Omega = Ce^{-\mu\sigma H\beta} = Ce^{\beta E(\sigma)}$$

$$P = \frac{1}{\Omega} = \frac{1}{Ce^{\beta E(\sigma)}} = Ce^{-\beta E(\sigma)}$$

C.1)

$$P(\sigma_1, \sigma_2) = P(\sigma_1) P(\sigma_2)$$

- $P(\sigma_1) = A e^{-\beta H \mu \sigma_1}$
- $P(\sigma_2) = B e^{\beta H \mu \sigma_2}$

$$C = AB$$

$$P(\sigma_1, \sigma_2) = C e^{\beta H \mu (\sigma_1 + \sigma_2)}$$

- $\langle X \rangle = \sum_j x_j P_j$

$\langle \sigma_1 \sigma_2 \rangle =$

# Prob 2.)

a.) $Q(\beta) = \frac{1}{h}\int dp\,dq\, e^{-\beta H(p,q)}$ **objective:** • Calculate the partition function.

• $H(p,q) = \frac{p^2}{2m} + \frac{Kq^2}{2}$

$\left(\beta = \frac{1}{K_B}\right.$ $\quad -\infty < p < \infty$ $\quad -\infty < q < \infty$

$$Q(\beta) = \frac{1}{h}\int_{-\infty}^{\infty} dp \int_{-\infty}^{\infty} dq\, e^{\frac{-\beta p^2}{2m}} e^{\frac{-\beta K q^2}{2}}$$

$$\int_{-\infty}^{\infty} e^{-ax^2} dx = \sqrt{\frac{\pi}{a}}$$

$$Q(\beta) = \frac{1}{h}\int_{-\infty}^{\infty} e^{\frac{-\beta}{2m}p^2} dp \int_{-\infty}^{\infty} e^{\frac{-K\beta}{2}q^2} dq \quad \left(\text{from tables \& Notes}\right)$$

$a = \frac{\beta}{2m}$ $\qquad$ $a = \frac{K\beta}{2}$

$$Q(\beta) = \frac{1}{h}\sqrt{\pi\left(\frac{2m}{\beta}\right)}\sqrt{\pi\left(\frac{2}{K\beta}\right)}$$

$$Q(\beta) = \sqrt{\frac{\pi^2 4m}{K\beta^2 h^2}} = \frac{2\pi}{\beta h}\sqrt{\frac{m}{K}}$$

- $\omega = \sqrt{\frac{k}{m}}$

- $\boxed{Q(\beta) = \frac{2\pi}{\beta h \omega}}$

b.) $\underline{\langle E \rangle}$ :

$$\langle E^2 \rangle = -\left(\frac{\partial \ln(Q)}{\partial \beta}\right)_{N,V}$$

- $\beta = \frac{1}{k_B T}$

$$\ln\left(\frac{2\pi}{\beta h \omega}\right) = \underbrace{\ln\left(\frac{2\pi}{h\omega}\right)} - \ln(\beta)$$

- This is a constant
  call it A

$$\ln\left(\frac{2\pi}{\beta h \omega}\right) = A - \ln(\beta) \qquad \boxed{\langle E \rangle = k_B T}$$

$$\langle E \rangle = -\frac{\partial}{\partial \beta}\left(A - \ln(\beta)\right) =$$

$$0 + \frac{1}{\beta} = k_B T$$

$$\langle (\delta E)^2 \rangle = \langle E^2 \rangle - \langle E \rangle^2$$

$$= - \left( \frac{\partial \langle E \rangle}{\partial \beta} \right)_{N,V}$$

- $\langle E \rangle = \frac{1}{\beta}$

$$\langle (\delta E)^2 \rangle = - \frac{\partial}{\partial \beta} \left( \frac{1}{\beta} \right)$$

$$= - \left( -\beta^{-2} \right) = \frac{1}{\beta^2} = k_B^2 T^2$$

$$\boxed{\langle (\delta E)^2 \rangle = k_B^2 T^2}$$

$$\langle (\delta E)^2 \rangle = k_B T^2 C_V$$

$$k_B^2 T^2 = k_B T^2 C_V$$

$$\boxed{C_V = k_B}$$

$$A(T) = E - TS = -\frac{\ln Q}{\beta}$$

$$\boxed{A = -\frac{\ln}{\beta} \left( \frac{2\pi}{\beta \hbar \omega} \right)}$$

- $\beta = \frac{1}{k_B T}$

- $S = -\left(\frac{\partial A}{\partial T}\right)_{V,N}$

- $\boxed{A(T) = -k_B T \ln\left(\frac{2\pi k_B T}{\hbar\omega}\right)}$

- $S = -\frac{\partial}{\partial T}\left(-k_B T \ln\left(\frac{2\pi k_B T}{\hbar\omega}\right)\right)$

$$= k_B T \frac{\partial}{\partial T}\left(\ln\left(\frac{2\pi k_B T}{\hbar\omega}\right)\right) + k_B \ln\left(\frac{2\pi k_B T}{\hbar\omega}\right)\frac{\partial}{\partial T}(T)$$

$$= k_B T \left(\frac{\hbar\omega}{2\pi k_B T}\right)\left(\frac{2\pi k_B}{\hbar\omega}\right) + k_B \ln\left(\frac{2\pi k_B T}{\hbar\omega}\right)$$

$$\boxed{S = k_B\left(1 + \ln\left(\frac{2\pi k_B T}{\hbar\omega}\right)\right)}$$

c.)

## Equipartition theory:

- Energy is shared equally amongst all energetically accessible degrees of freedom.

- each quadratic degree of freedom posseses a $\frac{1}{2}kT$ term.

- The Hamiltonian

$$H = \frac{p^2}{2m} + \frac{kq^2}{2}$$

- since q, and p variables have quadratic terms both potential and kinetic Energy contribute $\frac{1}{2}kT$

- $\langle H \rangle = \frac{1}{2}kT + \frac{1}{2}kT = kT = \langle E \rangle$
- Should get the same expectation value of energy.

d.)

- Since I knew are partition function I can then multiply it 3N times.

$$Q_{new}(\beta) = [Q(\beta)]^{3N}$$

$$\boxed{Q_{new}(\beta) = \left(\frac{2\pi}{\beta h w}\right)^{3N}}$$

- $\langle E \rangle = -\left(\frac{\partial \ln(Q)}{\partial \beta}\right)_{N,V}$

- $= -\frac{\partial}{\partial \beta}\left(3N \ln\left(\frac{2\pi}{\beta h w}\right)\right)$

$= -\frac{\partial}{\partial \beta}\left(3N \ln\left(\frac{2\pi}{hw}\right) - 3N \ln(\beta)\right)$

$= -\left(-3N\right)\left(\frac{1}{\beta}\right) = 3N K_B T$

$$\boxed{\langle E \rangle = 3N K_B T}$$

- $\langle (\delta E)^2 \rangle = -\left(\frac{\partial \langle E \rangle}{\partial \beta}\right)_{N,V}$

- $\langle E \rangle = \frac{3N}{\beta}$

- $\langle (\delta E)^2 \rangle = -\frac{\partial}{\partial \beta}\left(\frac{3N}{\beta}\right) =$

$$= -3N\left(\frac{-1}{\beta^2}\right) = \frac{3N}{\beta^2}$$

$$= 3N k_B^2 T^2$$

- $\boxed{\langle (\delta E)^2 \rangle = 3N k_B^2 T^2}$

- $A(T,N) = \frac{-\ln(Q)}{\beta}$

- $\ln(Q) = 3N \ln\left(\frac{2\pi k_B T}{h\omega}\right)$

$$\boxed{A(T,N) = -3N k_B T \ln\left(\frac{2\pi k_B T}{h\omega}\right)}$$

- $S = -\left(\frac{\partial A}{\partial T}\right)_{V,N}$

- $S = -\frac{\partial}{\partial T}\left[-3Nk_BT\ln\left(\frac{2\pi k_B}{\hbar\omega}\right) - 3Nk_BT\ln\left(\frac{T}{\quad}\right)\right]$

- $S = -\left(-3Nk_B\ln\left(\frac{2\pi k_B}{\hbar\omega}\right) - 3Nk_B\ln\left(T\right)T - 3Nk_BT\left(\frac{1}{T}\right)\right)$

- $\boxed{S = 3Nk_B\left[\ln\left(\frac{2\pi k_BT}{\hbar\omega}\right) + 1\right]}$

$$C(T,N) = \frac{\langle(\delta E)^2\rangle}{k_BT^2}$$

$$C = \frac{3Nk_B^{\not 2}\,\not T^2}{k_B\not T^2}$$

$$\boxed{C_v = 3Nk_B}$$

Prob 3.)

a.) $Q(T,N)$: (1-D oscillator)

- $Q(\beta, N, V) = \sum_{r} e^{-\beta E_r}$

- $E_n = \hbar\omega\left(n + \frac{1}{2}\right)$, $n = 0, 1, 2, \ldots$

$$Q(T,N) = \sum_{n=0}^{\infty} e^{-\beta\hbar\omega\left(n+1/2\right)}$$

$$= e^{-\frac{\beta\hbar\omega}{2}} \sum_{n=0}^{\infty} \left(e^{-\beta\hbar\omega}\right)^n$$

Geometric series:

$$\sum_{n=0}^{\infty} r^n = \frac{1}{1-r}$$

- $r = e^{-\beta\hbar\omega}$

$$Q(T,V) = \frac{e^{-\frac{\beta\hbar\omega}{2}}}{1 - e^{-\beta\hbar\omega}} \left(\frac{e^{\frac{\beta\hbar\omega}{2}}}{e^{\frac{\beta\hbar\omega}{2}}}\right)$$

$$Q(T,N) = \frac{1}{\exp\left(\frac{\beta\hbar\omega}{2}\right) - \exp\left(\frac{-\beta\hbar\omega}{2}\right)}$$

$$Q(T,N) = \frac{1}{2\sinh\left(\frac{\beta\hbar\omega}{2}\right)}$$

1-D

3-D:

$$Q(T,N) = \left(\frac{1}{2\sinh\left(\frac{\beta\hbar\omega}{2}\right)}\right)^{3N}$$

- $\beta = \frac{1}{K_B T}$

## A(T,N):

- $-\beta A = \ln(Q)$
- $-\beta A = \ln\left(\frac{1}{2\sinh\left(\frac{\beta\hbar\omega}{2}\right)}\right)^{3N}$

$$A(T,N) = -3N K_B T \ln\left(\frac{1}{2\sinh(\hbar\omega/2K_BT)}\right)$$

- $S = -\left(\frac{dA}{dT}\right)_{N,V}$

- $A(T,N) = -3N k_B T \ln\left(\frac{1}{2\sinh(\hbar\omega/2k_BT)}\right)$

- $A(T,N) = -3Nk_BT\left[\ln(1) - \ln\left(2\sinh\left(\frac{\hbar\omega}{2k_BT}\right)\right)\right]$

- $A(T,N) = 3Nk_BT \ln\left(2\sinh\left(\frac{\hbar\omega}{2k_BT}\right)\right)$

- $S = -3Nk_B\left[\frac{\partial}{\partial T}(T)\ln\left(2\sinh\left(\frac{\hbar\omega}{2k_BT}\right)\right)\right.$

$\left. + T\frac{\partial}{\partial T}\left(\ln\left(2\sinh\left(\frac{\hbar\omega}{2k_BT}\right)\right)\right)\right]$

$S = -3Nk_B\left[\ln\left(2\sinh\left(\frac{\hbar\omega}{2k_BT}\right)\right)\right.$

$\left. + T\left(\frac{1}{2\sinh\left(\frac{\hbar\omega}{2k_BT}\right)}\right)\left(2\cosh\left(\frac{\hbar\omega}{2k_BT}\right)\right)\left(\frac{-\hbar\omega}{2k_BT^2}\right)\right]$

$$S = -3Nk_B \ln\left(2\sinh\left(\frac{\hbar\omega}{2k_BT}\right)\right)$$

$$+ \frac{3Nk_B\hbar\omega T}{2k_BT^2} \frac{2\cosh\left(\hbar\omega/2k_BT\right)}{2\sinh\left(\frac{\hbar\omega}{2k_BT}\right)}$$

$$\boxed{S = -3Nk_B \ln\left(2\sinh\left(\frac{\hbar\omega}{2k_BT}\right)\right) + \frac{3N\hbar\omega}{2T}\coth\left(\frac{\hbar\omega}{2k_BT}\right)}$$

## $E(T, N)$ :

Fundamental equation :

- $A = E - ST$ ; $A + ST = E$
- $ST = -3NTk_B \ln\left(2\sinh\left(\frac{\hbar\omega}{2k_BT}\right)\right) + \frac{3N\hbar\omega}{2}\coth\left(\frac{\hbar\omega}{2k_BT}\right)$
- $A = -3NK_BT\ln\left(\frac{1}{2\sinh\left(\frac{\hbar\omega}{2k_BT}\right)}\right)$

- $E(T,N) = -3Nk_BT \ln\left(\frac{1}{2\sinh(\hbar\omega/2k_BT)}\right)$

$$-3NTk_B \ln\left(2\sinh\left(\frac{\hbar\omega}{2k_BT}\right)\right) + \frac{3N\hbar\omega}{2}\coth\left(\frac{\hbar\omega}{2k_BT}\right)$$

- $E(T,N) = -3NTk_B\left[\ln\left(\frac{1}{2\sinh\left(\frac{\hbar\omega}{2k_BT}\right)}\right)\right] + \frac{3N\hbar\omega}{2}\coth\left(\frac{\hbar\omega}{2k_BT}\right)$

- $\ln(x) - \ln\left(\frac{1}{x}\right) = \ln(x^2)$

- $E(T,N) = 3NTk_B \ln\left(\sinh\left(\frac{\hbar\omega}{2k_BT}\right)\right) + \frac{3N\hbar\omega}{2}\coth\left(\frac{\hbar\omega}{2k_BT}\right)$

$$E(T,N) = \frac{3N\hbar\omega}{2}\coth\left(\frac{\hbar\omega}{2k_BT}\right)$$

# C(T).

- $C(T,N) = \left(\frac{\partial E}{\partial T}\right)_{V,N}$

$\underline{E(T,N)}$: $\frac{3N\hbar\omega}{2}\coth\left(\frac{\hbar\omega}{2k_BT}\right)$

- $\frac{d}{dx}(\coth u) = \frac{-1}{\sinh^2 u}\frac{du}{dx}$

- $C(T,N) = \frac{-3N\hbar\omega}{2\sinh^2\left(\frac{\hbar\omega}{2k_BT}\right)}\left(\frac{\hbar\omega}{2k_B}\right)\left(\frac{-1}{T^2}\right)$

$$C(T,N) = \frac{3N\hbar^2\omega^2}{4k_BT^2\sinh^2\left(\frac{\hbar\omega}{2k_BT}\right)}$$

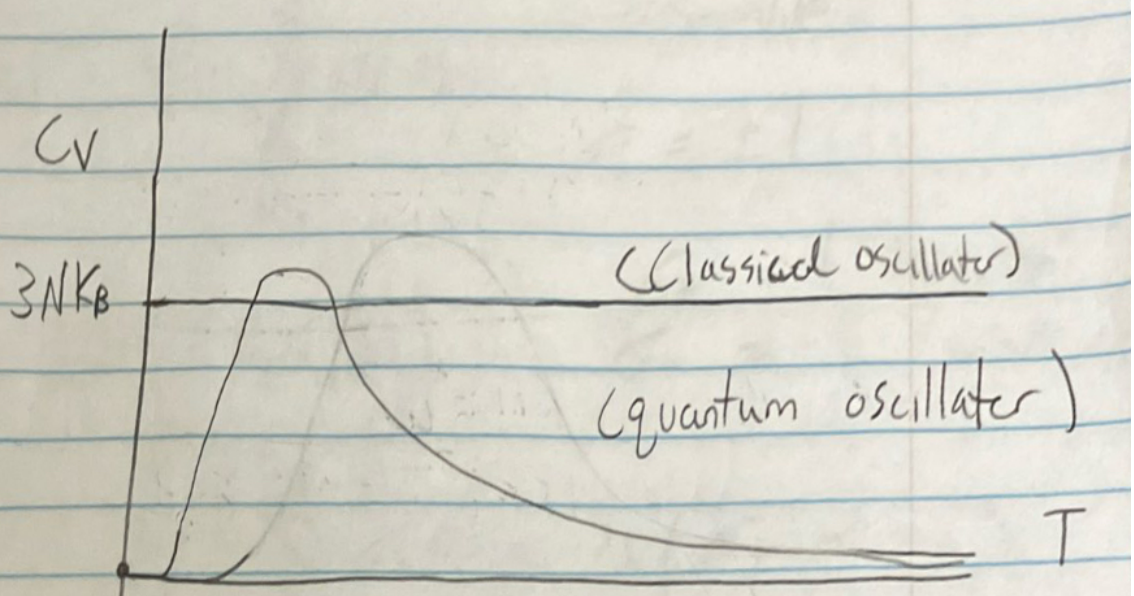
b.)

## Classical oscillator:

$$C_V = 3NK_B$$

## Quantum oscillator:

$$C_V = \frac{3N\hbar^2\omega^2}{4k_BT^2\sinh^2\left(\frac{\hbar\omega}{2k_BT}\right)}$$




- Results by looking at Desmos

- for a classical oscillator the heat capacity will be the same for all temperatures in comparison for a quantum oscillator the heat capacity has maxima and then at higher temps the limit will reach zero.